

Criterio cauchy

Teorema 1 (Criterio de Cauchy). *Sea $\sum_n z_n \subset \mathbb{C}$ una serie, entonces*

$$\sum_n z_n \text{ converge} \iff \forall \varepsilon > 0 : \exists N_0 \in \mathbb{N} : \forall N \geq N_0, k \in \mathbb{N} : \left| \sum_{n=N}^{N+k} z_n \right| < \varepsilon.$$

Demostración: $\sum_n z_n \text{ converge} \iff (s_n)_{n \in \mathbb{N}} \text{ converge} \iff (s_n)_{n \in \mathbb{N}} \text{ es de Cauchy}$

$$\iff \forall \varepsilon > 0 : \exists N_0 \in \mathbb{N} : \forall n, m = n + k \geq N_0 : |s_n - s_m| = \left| \sum_{k=n}^{n+k} z_k \right| < \varepsilon.$$

■